VIDHYASAATHY

SMALL STEPS MAKE BIG PROGRESS



THE PRODUCT MASTER FILE

VidhyaSaathy

The definitive document that decides how everything works — from the first sign-in to a fully working, adaptive learning app for Indian schools.

4 Roles · 4 Worlds | NCERT Class 9-10 | 4-Day Build Plan | Full Readiness Checklist | English Edition

> This file is the single source of truth for the product. Every feature — quizzes, flashcards, school notes, interactive notes, summary PDFs, curated YouTube, games, the AI cloud-mascot chat, parent dashboards and the four-band adaptive engine — is defined here with its exact flow, the data behind it, and the checklist item that proves it works. If a screen or rule is not in this file, it does not get built.

DOCUMENT	VERSION	SCOPE	BUILD WINDOW
Product Master File	1.0 — Definitive	Prototype · NCERT 9 & 10	4 Days

COMPANION

Build Prompts 01–10

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What This File Decides

Twelve parts, in the order you will need them: what the product is, who it serves, how the engine works, how every feature flows, and exactly what must be true by the end of Day 4. Page numbers refer to this PDF.

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+ **170-item master tracker** Part 10



How to read the numbers: page numbers refer to body numbering — "1" is the first content page (How to Use This File). The cover and this contents page are unnumbered. If a screen, rule or flow is ever missing from this file, it does not get built; that is the whole point of a master file.

ORIENTATION

How to Use This File

Read this once before anything else. It explains what this document is, what power it has, and how it connects to the Build Prompts you will paste into Cursor or Claude Code.

This is the **Product Master File** for VidhyaSaathy. It is the document your backend decisions are checked against: every table, every rule, every screen flow and every acceptance test in the working app must trace back to a line in this file. It replaces the current situation — features exist as beautiful static screens, but nobody has defined how the app actually flows from sign-in to learning to progress — with one definitive, deliberate answer for every part of the product.

The file is written in plain English on purpose. It is meant to be read by three audiences at once: you (the founding team, making product calls), the AI builder (Cursor or Claude Code, which will receive the companion Build Prompt files), and any future engineer or teacher who joins and needs to understand the whole system in one sitting. Where a rule is a hard rule, it is marked as such; where something is deliberately deferred to after launch, this file says so explicitly so nothing silently creeps into the four-day build.

> **The relationship to the Build Prompts:** the folder **VidhyaSaathy_Build_Prompts** contains one master prompt plus ten phase prompts (01 Database → 10 Launch) that instruct the AI builder. Those prompts say *how to build*. This file says *what the product is and how it must behave*. When the two ever disagree, this file wins — update the prompt, not the product truth. Before pasting any phase prompt, its matching part here should be re-read.

What this file fixes forever

The complete flow from sign-up and sign-in to learning, practice, games, AI help and parent reports; the four performance bands and how each child is adaptively served; every feature's data and flow; the NCERT 9-10 content scope; the four-day execution plan; and the full readiness checklist you will tick off item by item.

What this file deliberately does not do

It does not include public leaderboards, coaching-institute integrations, fee or transport modules, camera-based emotion detection, or AR/VR — these are permanent cuts from the product strategy, recorded so they are never "accidentally" built. Deferred signature features (the advanced AI suite, PIN profiles, recordings pipeline) sit in the post-launch backlog in Part 12, in priority order.

4
ROLES — STUDENT, TEACHER, PARENT, ADMIN

2
CLASSES — NCERT 9 & 10 ONLY

4
PERFORMANCE BANDS ROUTING EVERY CHILD

5
GAMES ON ONE SHARED QUESTION BANK

4
DAYS FROM REPO AUDIT TO LAUNCHED APP

One final orientation note on tone. VidhyaSaathy is built around a simple promise made to a tired Class-9 student: *"You don't need to decide what to study tonight. The app already knows, it knows how you learn, and it will walk with you."* Every section that follows is just that promise, made engineering-real — the diagnostic that finds what you actually know, the bands that route you to what actually helps, the cloud companion that removes the chaos, and the evidence trail that proves to your teacher and your parents that you are moving. Keep that promise in mind whenever a decision feels ambiguous; it settles most arguments.

PART 1

Product Charter

What VidhyaSaathy is, the problem it kills, why the app itself — not any single feature — is the moat, and exactly what "ready" means at the end of the four-day build.

1.1 — What VidhyaSaathy Is

VidhyaSaathy is a school-anchored learning companion for Indian students of Class 9 and 10, built on NCERT content, in which the school's own structure — classes, sections, roll numbers, subjects, teachers — is the backbone of the product. The app mirrors the school instead of replacing it: teachers publish what they taught, students get one clear, adaptive path through it, parents finally see real progress in plain language, and the admin runs the whole world from one console. Four roles, one shared truth: every piece of content, every quiz attempt, every doubt and every plan lives against the same concept map, so evidence flows from the student's screen to the teacher's heatmap to the parent's weekly summary without anyone re-entering anything.

The product's engine is deliberately simple and deliberately honest: **diagnose first, teach second, adapt always.** Before a student is taught anything, a short diagnostic identifies what they already know; the four performance bands (0-25%, 25-50%, 50-75%, 75-100%) then route them to the explanation depth, practice difficulty and pacing that fits; and as they work, every answer — including how confident they felt answering it — feeds back into per-concept mastery scores that re-route the next session. Nothing in the app is decorative: every screen either teaches, tests, adapts, reports, or removes chaos.

1.2 — The Problem We Kill

The enemy is not ignorance; it is **chaos**. A Class-9 student preparing for a Physics test today is lost between a YouTube video of unknown quality, a PDF of someone else's notes, three WhatsApp groups, a friend's photocopy, and a coaching institute's timetable — with no idea which one matches what *their* teacher taught, *their* syllabus, and *their* actual gaps. Parents cannot help because they cannot see inside the chaos; they only see marks. Teachers cannot help after 5 pm because they have no view of what each child actually struggled with at home.

> "The problem was never that the student is weak. The problem is that 17 different sources are shouting at once and none of them knows the student."
>
> FOUNDING INSIGHT — FROM THE ORIGINAL PRODUCT VISION

VidhyaSaathy's answer is structural, not cosmetic: one Next-Action card that tells the student the single right thing to do tonight, one curated set of resources that the teacher approved for this exact concept, one cloud companion that answers doubts at 11 pm without judgement, and one progress trail that turns effort into visible proof for parents and teachers. Removing the chaos — not adding more content — is the product's core value, and it is why the home screen will always be quieter than any competitor's.

1.3 — The Moat Statement

Features are copyable; the loop is not. Any competitor can ship quizzes, flashcards and an AI chatbot in a quarter. What they cannot copy is the compounding data flywheel that this app is designed around from day one, inside one real school, on one real board (CBSE/NCERT), with teachers publishing daily and students practising nightly:

FLYWHEEL 1

Every answer is evidence

Correctness, confidence, time-to-answer and error type logged per concept, per child, every day.

FLYWHEEL 2

Evidence improves routing

Band placement, difficulty ladders, revision radar and reteach heatmaps get sharper each week.

FLYWHEEL 3

Routing improves outcomes

Students feel the app "gets" them — the personal-tuition feeling at school scale.

FLYWHEEL 4

Outcomes earn trust

Teachers keep publishing, parents keep linking, students keep showing up nightly.

FLYWHEEL 5

Trust deepens the data

More nightly evidence → the loop tightens and the moat widens, school by school.

The strategic consequence is a strict priority order for the four-day build: the evidence trail (Part 3) and the adaptive routing (Part 4) are engineered **before** any delight feature, because the moat compounds on data quality, not on animation polish. A beautiful game that does not write concept-level evidence is a toy; a plain quiz that does is an asset.

1.4 — Prototype Scope — NCERT Class 9 & 10

The entire prototype is trained and seeded on NCERT Class 9 and Class 10 content only, for one demo school with two sections per class. This is a permanent scope decision for the prototype phase, recorded here so the build never wanders into other boards, classes or languages. The demo anchor is **Class 9-A, Physics, Chapter "Motion"**, with a three-concept prerequisite chain (distance & displacement → speed & average speed → acceleration), because it exercises every mechanic in the product: prerequisites, diagnostics, band routing, confidence signals, error DNA, reteach heatmaps and parent summaries.

Scope dimension	In the prototype	Explicitly out (for now)
Curriculum	NCERT Science + Mathematics, Class 9 & 10	Other boards (ICSE/State), other classes, other subjects
School model	One school, two sections per class, real roll numbers	Multi-school tenancy, coaching institutes
Languages	English + Hindi (full parity), Hinglish tone in AI chat	Other Indian languages
AI surface	Cloud-mascot chat, SnapDoubt, Socratic homework mode, plan maker	Voice interaction, emotion detection (cut forever), AR/VR
Devices	Low-end Android phones via PWA, offline-first sessions	Native app store builds

1.5 — What "Ready in 4 Days" Means

"Ready" is defined before building begins, and it is deliberately a product definition, not a technology one. At the end of Day 4 the following sentence must be true end-to-end, demonstrated live on a real phone: *a school admin has set up the school and issued activation codes; a teacher has published tonight's session for Class 9-A with a diagnostic, notes and a quiz; three different students — strong, struggling and absent — have signed in, been routed differently by their bands, completed the session, asked the cloud companion a doubt, and played one game; and a parent has opened a dashboard with real graphs generated from those exact events.* Every checklist item in Part 10 exists to make that one sentence true, and no work that does not serve it belongs in the four days.

> **Non-negotiable quality bar:** "ready" means **working flows with real data** — server-side scoring, real database reads and writes, row-level security on, no placeholder JSON, no mock logins. Screens that merely look finished do not count; Part 11's acceptance script will test behaviour, not pixels.

PART 2

The Four Worlds — Users & Roles

VidhyaSaathy is four products sharing one truth. This part defines what each role wants, what they get, the screens they live on, and who may touch whose data.

2.1 — Role Cards

The Student — "Just tell me what to do tonight"

A 14-15 year old with a low-end Android phone, 30-60 free minutes on a school night, and exam pressure. The student wants zero decisions: one next action, explanations at their level, help without judgement at 11 pm, and small wins that feel earned.

Gets: adaptive home with Next-Action card, diagnostic before new chapters, interactive concept pages, quizzes with confidence buttons, flashcards, summary PDFs, curated YouTube, five games, streaks and badges, the cloud companion for doubts and plans.
Never sees: their band label, other students' scores, public ranking.

The Teacher — "40 students, 5 minutes per evening"

The subject teacher (Sharma Ma'am, Physics) who already works hard in class and has no evening bandwidth. The teacher wants to publish tonight's session in under three minutes, see exactly where the class stumbled, and reteach one concept with one tap instead of guessing.

Gets: morning dashboard, class × concept heatmap with drill-down, session builder with student-preview, question bank with auto-selection, short-answer grading queue, anonymous doubt clusters ("17 doubts on acceleration").
Never does: manual data entry of student progress — evidence arrives on its own.

The Parent — "Show me progress I can act on"

A working parent who currently judges learning only by exam marks, weeks after the fact. The parent wants a truthful weekly picture in plain language, one concrete conversation to have, and reassurance without surveillance.

Gets: weekly summary dashboard with graphs (effort, mastery trend, streak), "Ask this" conversation prompts, milestone notifications, printable report.
Never sees: the child's live AI chats, doubts, or raw answers — a hard privacy boundary (2.3).

The Admin — "Run the school's digital world safely"

The school's coordinator who owns setup and accountability. The admin wants bulk onboarding without chaos, control over who sees what, and an audit trail when anything goes wrong.

Gets: school setup wizard, CSV import of students and staff, activation-code generation and export, parent-link approvals, permission matrix, audit events, demo-reset script.
Never touches: students' AI conversations or doubt content (same privacy wall).

2.2 — The School-Anchored Identity Model

Nobody "creates an account" in the casual consumer sense. The app is a mirror of the school's real structure, so identity flows from it: the admin creates the school, academic year, classes and sections; teachers are invited by staff ID and email; students activate with a code plus their roll number and date of birth; parents are linked to their child via an OTP after school approval. This anchoring is what makes every later feature trustworthy — a "Class 9-A Physics session" means a real teacher's real session for a real section, not a generic course. It is also the security backbone: class membership, subject assignments and parent-child links are database facts that row-level security checks on every request, so a parent can never read another child's data and a student can never read answer keys.

2.3 — Screen Maps per Role

Role	Primary screens (in daily order of use)	Navigation shell
Student	Adaptive Home (Next-Action) · Class Workspace · Concept Page (notes + checkpoints) · Quiz Player · Session Summary ("Why I got it wrong") · Doubt Chat with cloud companion · Games Arcade · Flashcards & Summary Sheets · Profile (streak, badges, mastery map)	Bottom tab bar — 5 tabs, thumb-first

Teacher	Morning Dashboard · Class × Concept Heatmap (drill + reteach) · Session Builder (with student preview) · Question Bank · Grading Queue · Doubt Clusters · Class Space (announcements)	Left sidebar (desktop) / top-drawer (mobile)
Parent	Weekly Home (graphs + plain-language summary) · Progress Detail per subject · "Ask This" card · Notifications · Report (print/share)	Simple top-nav, 3 entries maximum
Admin	School Setup · People (students/staff/parents) · Activation Codes · Approvals · Permission Matrix · Audit Log · Demo Reset	Left sidebar with admin-only sections

2.4 — Role × Feature Permission Matrix

The matrix below is the security and product truth for who can do what. It is enforced server-side by row-level security policies, never merely hidden in the UI; the build prompts implement exactly this table.

Capability	Student	Teacher	Parent	Admin	Hard boundary
Read published content & sessions	Own class	Own classes	—	All	Section-scoped for students
Attempt quizzes / games	Yes	—	—	—	Server-side scoring only
See answer keys	Never	Yes	—	—	Keys never reach the student client
Read child's progress	Own	Own students	Linked child	All	Aggregates only for parents
Read AI chats / doubts / raw answers	Own	Anonymised clusters	Never	Never	Privacy wall — no exceptions
Publish sessions & content	—	Yes	—	—	Own subject & section only
Approve AI-drafted questions	—	Yes	—	—	AI drafts never auto-publish
Manage people, codes, links	—	—	—	Yes	Audit-logged actions

> **Why the privacy wall matters commercially:** students will only confess real confusion to the cloud companion if they are certain no parent reads it. The wall is not compliance theatre — it is what makes the doubt data (and therefore the teacher's heatmap) honest.

PART 3

The Core Product Engine

One loop runs the entire app — Diagnose, Learn, Practice, Understand, Progress. Every feature exists to serve one station of this loop, and every station writes evidence the next station consumes.

3.1 — The Loop

STATION 1
Diagnose
A short, low-stakes test before teaching anything new — find what the student already knows and which prerequisites are shaky.

STATION 2
Learn
Concept pages at the student's band depth: school notes, interactive checkpoints, curated video, worked examples.

STATION 3
Practice
Adaptive quizzes and games on the same concept bank — difficulty climbs only as fast as the student proves it should.

STATION 4
Understand
"Why I got it wrong" feedback: concept tag + error type + the one fix. Doubts go to the cloud companion instantly.

STATION 5
Progress
Per-concept mastery updates, streaks and badges for the student, heatmaps for the teacher, weekly graphs for the parent.

Progress feeds the next Diagnose — the loop never resets, so the app knows the student a little better every single night.

This loop is the product's spine and its scheduling discipline: no feature ships that does not sit at a named station. The games arcade, for example, lives at Station 3 (Practice) and Station 5 (Progress) — which is why every game scores against concepts and never against classmates. The flashcard deck lives at Station 2 and 3 as a Rapid Revision surface. The parent dashboard is Station 5 made visible. When a future idea arrives ("should we add a doubt marketplace?"), the first question is which station it serves; if the answer is none, it does not belong in VidhyaSaathy.

3.2 — Diagnostic-First Rule (Test Before Teach)

Before a student is taught any new chapter, they take a short diagnostic — roughly five questions drawn from the chapter's prerequisite chain plus a taste of the new material. The result places each concept on the student's mastery map and decides the entry band: the 0-25% student gets prerequisite repair first, while the 75-100% student is challenged immediately. The diagnostic is framed to the student as a friendly "let's see what you already know — this is not a test," it is never graded, and it is never shown to parents as marks. Its output is routing, not judgement.

This rule exists because teaching a student what they already know wastes their scarce 40 minutes and quietly insults them, while teaching over a broken prerequisite guarantees that tonight's effort evaporates by morning. The diagnostic is cheap (five questions, three minutes) and its payoff is the entire personalisation premise. **Every new chapter, for every student, starts with a diagnostic — no exceptions, including chapters the teacher is confident about.**

3.3 — The Next-Action Card (Chaos Removal, Made Literal)

The student home screen is dominated by a single card that answers the only question that matters at 8 pm: *"what should I do right now?"* The card is computed server-side from the evidence trail — an unfinished session outranks everything; a due-tomorrow session outranks revision; revision of a decaying concept outranks new content; new content outranks games. It shows the concept, the estimated time, and one tap takes the student into exactly that activity mid-state. The rest of the home screen stays deliberately sparse: streak chip, one encouragement line, and quiet access to the rest of the world. Competing products fill this space with feeds; VidhyaSaathy fills it with calm, because the product promise in Part 1 was "you don't need to decide."

3.4 — Mastery Levels per Concept (L1-L4)

Level	Meaning (internal)	How it is earned	What the student sees
L1 — Fragile	Seen it, cannot yet use it	Diagnostic partial, or first attempts mostly wrong	"Let's fix this" + repair path (prerequisites re-taught simply)
L2 — Steady	Handles basic questions reliably	≥60% quiz on the concept at L1-L2 difficulty	"Almost there" + guided practice
L3 — Strong	Solves application problems	Consistent correctness at L3 difficulty, confidence "Sure"	"Mastered" + unlock Boss Level and Dushman questions

L4 — Advanced	Can handle exam twists	Board-format questions correct under time pressure	"Top form" + challenge set and next chapter preview

Mastery is **per concept, never a single overall percentage** — "Aarav: Distance 82%, Acceleration 30%" is the honest shape of a student, and an overall "56%" would hide exactly the gap the app exists to find. Levels decay gently over time (the revision radar, 5.6), so "mastered in September" must be re-proved lightly before half-yearly exams — matching how memory actually works.

3.5 — The Evidence Trail

Every learning event is appended to one immutable log — quiz answers with confidence buttons and time-to-answer, game scores with concept tags, doubts asked, sessions completed, notes finished, streak touches. Nothing overwrites history; corrections are new events. This trail is what makes every downstream screen honest at zero extra teacher effort: the heatmap is an aggregation of the trail, the parent graphs are an aggregation of the trail, the band routing is a query over the trail, and the AI's context window is the recent slice of the trail. It is also the moat's raw material (Part 1.3) — which is why the trail is designed and built on Day 1, before any screen that displays it.

PART 4

The Four Student Bands

The heart of personalisation: four internal performance bands — 0-25%, 25-50%, 50-75%, 75-100% — that silently route every child to the explanation depth, difficulty, pacing and tone that fits them, and move them up the moment they earn it.

4.1 — Two Non-Negotiable Rules (Read First)

- **Bands are internal, forever.** A student never sees a band label, a percentage rank, or any wording that smells like one. The student-visible vocabulary is only: "Mastered", "Almost there", "Let's fix this", "Top form". No leaderboards exist anywhere in the product — this is a permanent strategic cut, not a missing feature.
- **People are never labelled — concepts are.** Bands are a routing mechanism over per-concept mastery, not a verdict on a child. The same student is routinely Band-4 on "Distance" and Band-1 on "Acceleration", and the app must be equally comfortable serving both in one session.

4.2 — The Band Routing Table (Definitive)

Band	Internal meaning	Explanation & content served	Practice entry	Pacing & AI tone	Games & extras
0-25%	Large gaps — prerequisites shaky	Simplest explanation with everyday analogies; prerequisite concepts re-taught first from the concept map; shortest sentences; Hindi/Hinglish support prominent	L1 difficulty only, tiny sets of 3-5 questions, heavy guided practice with worked examples shown before attempting	Slow, lots of small wins; companion tone extra warm, zero jargon, celebrates single correct answers	Dushman (personal enemy) questions unlocked early on repaired concepts; Boss Level locked until L2
25-50%	Basics partly clear	Standard notes + worked examples step-by-step; interactive checkpoints after each idea; flashcards pushed for retention	Start L2, drop to L1 instantly on two consecutive errors with an automatic bridge explanation	Medium; companion checks understanding with one gentle question per concept before moving on	All games except Boss Level; daily challenge at L1-L2 mix

50-75%	Basics clear, application weak	Full concept pages + application focus: case-based examples, numericals, "where students typically slip" callouts	L2-L3 mix; adaptive ladder climbs after 3 consecutive correct	Steady; companion shifts from explaining to quizzing ("you seem solid — try this twist")	All games; Dushman targets the student's own 3 weakest concepts; Boss Level unlocks at L3
75-100%	Consistent understanding	Concise advanced notes; board-format twists; next-concept preview so boredom never arrives	L3 default + board-format questions; time-pressured sets for exam simulation	Fast; companion tone is coach-like and brief, treats the student as near-peer	Boss Level and challenge sets are the main loop; interest mash-ups (same concept, favourite context) keep novelty alive

4.3 — How Band Placement Is Computed

Band placement is deliberately simple, explainable and per-concept: the diagnostic seeds an initial mastery score per concept; every subsequent quiz answer updates it (correct-at-high-difficulty counts more; "Guess" confidence discounts a correct answer heavily; "Sure" + wrong is treated as a real misconception, the most diagnostic signal in the app); a rolling window (last ~10 interactions per concept) smooths noise. The band for a chapter is then derived from its prerequisite-weighted average. There is no black-box model in the prototype — simple, honest arithmetic the team can inspect, explain to a teacher, and debug at 11 pm. Sophistication can come later; trust must exist from day one.

4.4 — Promotion, Demotion and the Difficulty Ladder

Every question carries a difficulty tag — L1 basic, L2 medium, L3 application — and the ladder rule is fixed product law: **a student starts at L2; three consecutive correct climb to L3; two consecutive wrong drop to L1, always with a bridge explanation so the drop is felt as help, not punishment.** Promotion between bands follows sustained evidence (e.g. L2-L3 correctness at ≥70% across the chapter's concepts plus "Sure" confidence on most), and every transition is written to the log so a teacher can see "Aarav moved 25-50 → 50-75 on Motion, Tuesday" — progress made visible is the product's favourite kind of motivation. Demotion after absences or decayed mastery is quiet and shame-free: the app simply serves bridge content and never announces a fall.

4.5 — Friendly Language Mapping (Student-Facing Words)

Internal state	Student sees	Parent/teacher see
Band 1 (0-25%)	"Let's fix this together" + repair path	"Needs foundation support — working on prerequisites"
Band 2 (25-50%)	"Almost there — one more step"	"Developing — basics partly clear, guided practice on"
Band 3 (50-75%)	"Good going — try this twist"	"Progressing — ready for application questions"
Band 4 (75-100%)	"Top form — challenge yourself"	"Consistent — advancing to next concept"
Mastery decayed	"Quick warm-up?" (2-minute refresher)	"Revision radar flagged 2 concepts fading"

4.6 — The Same Evening, Three Different Children

The bands are best understood by watching one night in Class 9-A after Sharma Ma'am publishes the Motion session at 14:00 (session opens 18:00, closes 08:00 next morning):

Aarav — strong (Band 4 on prerequisites)

Diagnostic confirms distance/speed are solid; he is routed straight to the acceleration concept page, concise variant. He answers the quiz's L3 numericals in six minutes, taps "Sure" on four of five, misses one twist on sign convention — the "Why I got it wrong" card tags it *method error, not concept gap*, shows the one-line fix, and queues a Dushman question for tomorrow. Total time: 14 minutes. He then unlocks the Boss Level preview for the chapter and plays MCQ Rush for four minutes purely for fun. The streak counter ticks to 12.

Priya — struggling (Band 1 on acceleration, Band 2 overall)

The diagnostic flags that her "average speed" prerequisite is fragile. Instead of the full session, her Next-Action card offers a 10-minute repair path: the simplest distance-time explanation with a bus-journey analogy, then two guided examples she completes with hints. Only then does the acceleration page open — in its simplest variant, Hindi labels on. Her quiz is five L1 questions; she gets three right, taps "Guess" on one of those, which the engine discounts. The companion's tone stays warm, celebrates the third correct answer explicitly, and schedules tomorrow's bridge practice. No red, no failure language, anywhere.

Rohan — absent for three days (recovery path)

He opens the app at 20:30 after missing school. The Next-Action card does not dump three days of backlog; it offers "Catch-up, 20 minutes" — the revision radar picks the two most decayed concepts from before his absence plus tonight's session in compact form. The copy is shame-free: "Good to see you. Let's take this in small steps." The evidence trail logs a recovery event (distinct from a fresh attempt), so his teacher's heatmap shows "recovered" rather than a hole — and Sharma Ma'am's evening takes exactly one glance to know he is back.

One session, three experiences, one database — this is the adaptive promise made concrete, and it is the acceptance demo's centrepiece (Part 11). If any of the three children would have an identical evening in the built app, personalisation has failed and the phase does not pass its checklist.

PART 5

The AI Brain — Understanding Each Child

How the app learns the student: what signals it collects, how it diagnoses why an answer went wrong, how it predicts the way each child best learns, and the safety rules the AI may never break.

5.1 — The Signals We Collect (All Passive, All Logged)

Signal	What it is	What it tells the engine
Correctness	Right/wrong per question, per concept, at which difficulty	The mastery backbone — L1-L4 per concept
Confidence button	"Sure / Unsure / Guess" tapped before reveal, on every question	Separates knowledge from luck; "Sure + wrong" = real misconception; "Guess + right" = discounted
Time-to-answer	Seconds from question shown to answer locked	Too fast + wrong = careless; too slow + right = fragile mastery needing reinforcement
Error type	Classified after each wrong answer (5.2)	Chooses the repair path — not all wrongs are equal

Session rhythm	When the student studies, session length, resumptions	Plans realistic nightly goals; detects 11 pm cramming and frustration patterns
Avoidance	Concepts repeatedly opened then skipped; chat opened, doubt typed then deleted	The "Darr ka map" — fear map of concepts quietly avoided
Revision behaviour	Flashcard flips, summary-sheet opens before quizzes	Learning-approach evidence: does this child consolidate by re-reading or by testing?

Every signal is passive — the student never fills a "learning style questionnaire", because self-reported styles are unreliable and add friction. The only active gesture is the confidence button, and it survives in the product because students genuinely enjoy tapping it; it feels like a game mechanic while being the single most honest signal in the system.

5.2 — Error DNA (Why the Answer Went Wrong)

After every wrong answer the engine classifies the error, because the four families need completely different repairs. Mislabelling them is how generic apps waste student time — and how this one refuses to.

Error family	Signature in the signals	Repair path served
Concept gap	"Sure" + wrong + slow; prerequisite mastery low	Bridge back: prerequisite concept re-taught in simplest variant, then re-test
Careless slip	Very fast + wrong, on a concept otherwise at L3+	One-line nudge, no re-teach; a "slow down on signs" micro-habit card
Method error	Right approach, wrong execution (unit, sign, arithmetic)	Worked example of the exact same pattern, step highlighted where it broke
Guess	"Guess" confidence regardless of outcome	Honest acknowledgement, short explanation, question re-queued in a different skin tomorrow

5.3 — How the AI Predicts the Way Each Child Learns

The prediction layer is intentionally legible: it observes which explanation formats correlate with this child's subsequent correctness and shifts serving weights over time. If Priya answers consistently better after *worked examples with everyday analogies*, examples rise in her variant mix and dense theory sinks; if Aarav retains better after *testing himself* (flashcards and Dushman questions) than after re-reading, his Next-Action leans practice-first. The weights start neutral, move slowly, and are always inspectable in the profile ("explains best with: examples and Hindi terms; consolidates by: self-testing"). The AI never claims a scientific learning-style theory; it just stops serving what demonstrably does not work for this child. Prediction here means **pattern memory with a paper trail**, and that is a feature — teachers and parents can ask "why did it show her that?" and get a real answer.

5.4 — The Cloud Companion (Mascot, Chat and Plan-Maker)

The cloud mascot is the AI's face and the student's constant, calm companion — the one place help always lives. It is a friendly blue cloud with a small book, present in the chat and in micro-reactions across the student world, with eight restrained behaviour states: idle (subtle float), greeting, thinking (while the AI processes), encouragement after meaningful progress, confusion-support when the student is stuck, celebration at milestones, resting while the student reads or solves, and error (something failed — never the student's fault). Hard rules inherited from the brand: the cloud never animates distractingly during question-solving, never flashes for attention, never uses emotional pressure to keep a child online, and honours reduced-motion device settings everywhere.

In chat, the companion answers doubts in the student's chosen language (English, Hindi, or natural Hinglish), asks one clarifying question when the doubt is vague, and answers at the student's band depth — the same doubt about acceleration gets an analogy-first answer for Priya and a formula-forward answer for Aarav. As plan-maker it does the chaos removal of Part 1 literally: "exam in 3 weeks, 6 concepts shaky, 25 minutes a night — here is the plan", generated from the evidence trail and always realistic (plans that assume a suddenly-perfect student are lies, and children know it). Every conversation is private to the student by the privacy wall of Part 2.4.

5.5 — AI Safety Rules (Non-Negotiable)

- **No fabricated citations.** The AI never invents book page numbers, links or references. Content anchors only to concepts and resources that exist in the school's own library.
- **The honest "I don't know."** When unsure, the companion says so plainly and offers the doubt-escalation path — a wrong-but-confident answer from a tutor destroys the product's core trust.
- **Homework mode is Socratic.** If the teacher flagged the session as homework, the AI never hands the direct answer; it guides with questions and hints, then encourages the student's own attempt.
- **AI-drafted content is visually separate and teacher-gated.** AI-drafted practice questions appear in a distinct "AI draft" style and never reach students without explicit teacher approval.
- **Keys stay server-side.** The student client never receives correct-answer flags for quizzes — scoring happens on the server, always.
- **Tone guardrails.** No fear, no shame, no comparison with other children — encoded in the companion's system prompt and spot-checked in QA.

5.6 — The Radars (Quiet Intelligence)

Revision / warm-up radar

Tracks mastery decay on the forgetting curve; a 2-minute warm-up surfaces before decay hurts ("Quick warm-up? Distance & displacement — fading").

Frustration radar

Detects rapid wrong answers, fast retries and session abandonment; the companion softens tone, shrinks the next step, and may suggest a break. Never a red screen.

Ratta (rote) detector

Spotlights correct answers with "Sure" confidence that collapse when the same concept is asked with changed context — memorised form, not understanding.

Avoidance radar ("Darr ka map")

Maps quietly-avoided concepts and gently re-introduces them inside games where the stakes feel low — courage by disguise.

5.7 — The 11-Step Server-Side Request Sequence

Every AI chat turn follows one fixed server-side sequence (Edge Function, keys never in the client): 1) auth check → 2) rate-limit + token budget check → 3) load student context slice from the evidence trail → 4) load concept map around the doubt → 5) detect language and band depth → 6) assemble system prompt with safety rules and tone profile → 7) stream the answer (SSE) with the companion's thinking state → 8) log the interaction privately → 9) update doubt-cluster counters (anonymised) → 10) write any promised follow-ups (Dushman queue, plan items) → 11) close budget accounting. The sequence is deliberately spelled out here because it is the difference between an AI feature and an AI liability: no step may be skipped, and none of it runs on the student's phone.

PART 6

Master Feature Catalog

Every feature of VidhyaSaathy in one place — what it is, where it lives, how its flow works, and the data behind it. Eleven groups, A to K. If a feature is not in this catalog, it is not in the prototype.

A — Identity & Onboarding

Sign-in / Sign-up (school-anchored) VS-01..08 Core

Lives: public landing + auth screens | Built: Day 1

Role-aware entry with one clean flow per role: teachers accept an emailed invite tied to their staff ID; students activate with a code from their class teacher plus roll number and date of birth, then set a password; parents link via an OTP code issued through the school. No free-floating "create account" exists — identity is always anchored to the school's real structure, which is what makes every later permission honest.

Flow: landing → role select → invite/activate/link → SSR session → role-based redirect to the right shell. **Data:** profiles, memberships, activation codes (single-use, expiring), parent-child links, audit events on every join. **Must include:** forgot password, wrong-code friendly errors, English/Hindi toggle on all auth screens.

Parent linking & approvals VS-08 Core

Lives: parent onboarding + admin approvals | Built: Day 3

A parent enters the code, verifies by OTP, and the link waits in the admin's approval queue until the school confirms the relationship. Approved links grant read-access to progress aggregates only — never chats, doubts or raw answers — and can be revoked at any time with one admin action that is fully audit-logged.

Flow: code → OTP → pending link → admin approve → parent world opens. **Data:** parent-child link table with status, approval history, revocation events.

B — Diagnostic & Placement

Chapter diagnostic (test-before-teach) VS-19 Core

Lives: first open of any new chapter | Built: Day 2

~5 friendly questions from the chapter's prerequisite chain plus a taste of the new material; results seed per-concept mastery and decide the entry band. Framed as "let's see what you already know", never graded, never shown as marks to anyone. Skipping is allowed but the app simply defaults to Band-2 content until evidence arrives — the diagnostic remains the fastest path to a personalised evening.

Flow: chapter open → diagnostic offer → 5 questions with confidence buttons → placement result card (friendly copy) → routed home. **Data:** diagnostic attempts, per-concept seed scores, band assignment event in the log.

C — Learning Content (the five study surfaces)

School notes (teacher-authored) VS-11 Core

Lives: concept page | Built: Day 2

The teacher's own notes for the concept, written or pasted in the session builder, tagged to concepts before publish. This is the canonical text — what the teacher actually taught in class, in their own emphasis — so revision finally matches the classroom instead of a stranger's YouTube summary. Notes support simple formatting, images, and Hindi text natively.

Flow: teacher writes/pastes → concept tags required → publish → student sees notes on the concept page at their band variant. **Data:** notes table (subject/chapter/concept tags, versions, publish state).

Interactive notes VS-11+ Core

Lives: concept page | Built: Day 2

School notes upgraded with structure the app understands: sections per idea, an inline checkpoint question after each idea (one tap, instantly checked server-side), key-term highlighting, and a "explain this again simpler" action on every section that swaps to the simple variant of the same idea. Reading is therefore never passive — the page quietly tests comprehension as the student goes, and each checkpoint is evidence.

Flow: read section → checkpoint → instant feedback + micro-explanation if wrong → next section. **Data:** structured note blocks with concept tags, checkpoint attempts as evidence events.

Summary notes as PDFs (chapter one-pagers) VS-44 Core-P1

Lives: chapter resources + offline library | Built: Day 2

A printable one-page-per-chapter summary sheet assembled from the teacher's notes: definitions, the 3-5 core formulas, the two classic exam patterns, and the chapter's concept map in miniature. Students use it for the morning-of-exam scan; the data-saver audience gets it as a clean PDF download rather than a heavy page render. AI may draft it, but it publishes only after teacher approval, with the AI-draft visual marker removed once approved.

Flow: notes exist → generate draft (AI or template) → teacher approves → PDF in chapter resources → download/print/offline cache. **Data:** summary documents with approval trail and version history.

Flashcards (auto-built concept decks) VS-93+ Core

Lives: chapter resources + Rapid Revision game | Built: Day 4

Each concept automatically owns a small deck: front = term/question/mini-diagram, back = the answer in one breath, sourced from the teacher's notes and the question bank. Decks serve two moments: a deliberate 2-minute Rapid Revision before quizzes (which the engine recommends exactly when the revision radar flags decay), and free self-testing — the strongest known consolidation move. Card flips are logged as light evidence of revision behaviour.

Flow: chapter → deck → flip session (self-rated: knew it / almost / forgot) → feeds mastery signal. **Data:** cards per concept, flip + self-rating events.

Curated YouTube with timestamps VS-13 Core

Lives: concept page resources | Built: Day 2

YouTube is not the enemy — discovery is. The teacher pastes a link, sets start/end (e.g. "watch 8:20-15:40 for this concept"), tags the concept, and publishes with one line on why it is worth watching. The student gets a trimmed, contextualised, syllabus-matched video instead of an algorithmic rabbit hole. Watching is logged as engagement, never as mastery — the app is honest about what a view proves.

Flow: teacher pastes link → sets trim + concept + one-line reason → publish → student plays inline at timestamp. **Data:** resources table (type=video, timestamps, concept tag, reason).

D — Practice & Assessment

Quiz engine (server-scored) VS-18 Core

Lives: session + standalone practice | Built: Day 2

The workhorse of Station 3. Every quiz is drawn from the question bank by concept and difficulty, scored on the server (answer keys never reach the client), and ends in the "Why I got it wrong" summary — each miss with its concept tag and error family, each with its one-line fix. Sessions require ≥60% to complete; below that the app offers the bridge path rather than a bare "fail".

Flow: quiz start → per-question (answer + confidence button) → server score + error classification → summary with fixes → mastery update. **Data:** questions (difficulty, format, concept), attempts, per-question responses with confidence + timing.

Confidence buttons (Pakka Meter) VS-39 Core

Lives: every question, everywhere | Built: Day 2

One row of three buttons — Sure / Unsure / Guess — sits under every question in every quiz and game, tapped before revealing the answer. It is the app's single most diagnostic signal and doubles as a self-awareness trainer: students learn to notice the feeling of guessing, which is half the battle in board exams. Teachers get the class-level view ("12 students were 'Sure' and wrong on acceleration — that is a misconception, not carelessness").

Flow: answer selected → confidence tapped → reveal → engine weights the result. **Data:** confidence value on every response row.

Adaptive difficulty ladder VS-23 Core

Lives: inside every quiz and game | Built: Day 3

Fixed product law: start at L2; three consecutive correct climb to L3; two consecutive wrong drop to L1 with an automatic bridge explanation. The rule is simple enough to explain to a parent in one sentence and deterministic enough to debug — no magic, just a ladder that keeps every child in the zone where effort is visible.

Flow: engine picks next question by ladder state → transition logged → bridge text on any drop. **Data:** ladder state per student-concept, transition events.

Exam-format practice (Board mode) VS-24 P1

Lives: practice picker + test-prep signals | Built: Day 3

Every bank question carries a board-format tag — MCQ, Assertion-Reason, Case-Based, Numerical, Short Answer — so before exams the app serves "Board-format practice" on the concepts the student actually owns. Knowing a concept and performing it in exam clothing are different skills; this feature trains the second without pretending to be the first.

Flow: exam window or student choice → format-filtered set → timed → same feedback loop. **Data:** format tags on questions; format-wise accuracy in the trail.

Homework submission VS-17 Core

Lives: student homework list + teacher grading queue | Built: Day 3

Teachers assign homework attached to tonight's session; students submit short answers or attempts (text, and photo for written work in the prototype's simple form), and the submission lands in the teacher's grading queue with the student's evidence trail beside it. Objective items self-score; short answers queue for the teacher with AI-suggested marks that the teacher accepts or edits in one tap. In homework mode the companion is strictly Socratic (5.5).

Flow: assign → attempt (AI Socratic help) → submit → queue → grade → returned with one comment line. **Data:** assignments, submissions, grades, teacher edits to AI suggestions.

E — AI Features (all behind the cloud companion)

Doubt chat with the cloud companion VS-28 Core

Lives: always one tap away (chat tab) | Built: Day 3

Natural-language doubts at the student's hour of need — 11 pm included — answered at band depth, in English/Hindi/Hinglish, streaming token by token with the cloud's thinking state. The companion knows the student's recent evidence (privately) so its answers reference where the student actually is: "you slipped on the sign convention yesterday, watch it in this one". AMA-style general questions are allowed with truthful-source marking; homework-mode is Socratic; and every conversation is walled from parents and admins by the privacy rule.

Flow: chat open → doubt typed/voiced-to-text (prototype: typed) → 11-step server sequence (5.7) → streamed answer → private log + cluster counter. **Data:** conversations, messages, token spend, language detected.

SnapDoubt (photo to doubt) VS-30 P1

Lives: chat composer camera button | Built: Day 3

The classic 11 pm moment — a notebook page, one confusing line. The student photographs it, the app extracts the text server-side, shows it for confirmation ("is this the doubt?"), and only then sends. The confirmation step exists because mistranscription at midnight creates new confusion instead of killing old confusion; the student stays in control of what actually gets asked.

Flow: photo → extraction → confirm/edit → doubt chat continues as normal. **Data:** image in private storage, extracted text, the derived doubt.

Study plans (the chaos-killer) VS-34 Core

Lives: companion chat + profile "My plan" | Built: Day 3

"Exam in 3 weeks, 6 concepts shaky" → a realistic night-by-night plan generated from the evidence trail, honest about the student's actual available minutes, sequenced by prerequisite order so each night stands on the last. The plan adapts when life happens: a missed night shrinks the plan instead of breaking it. This is the feature that replaces the timetable-juggling stress students carry — and the one parents quote back most.

Flow: ask for plan (or exam auto-detected) → trail slice + calendar → plan card → nightly steps feed Next-Action. **Data:** plans, plan items, completion vs plan events.

Doubt clusters → teacher VS-38 Core

Lives: teacher doubt inbox | Built: Day 3

Doubts are clustered by concept and shown to the teacher anonymised — "17 doubts on acceleration, 9 of them on sign convention" — turning the invisible 11 pm confusion of a class into tomorrow's five-minute reteach. Named escalation is selective-in only: the student can choose "also tell my teacher", which files the doubt with their name into the teacher's inbox; the default is always anonymous. Teachers see clusters per concept with the actual anonymised snippets, not summaries of summaries.

Flow: doubts logged → nightly clustering job → teacher inbox card → optional one-tap reteach session push. **Data:** doubts, cluster assignments, escalation opt-ins, teacher actions.

F — Personalisation (beyond the bands)

Dushman questions (personal enemy) VS-41 Core

Lives: daily challenge + games | Built: Day 3

The three concepts that have personally defeated this student return, disguised, until they are beaten — re-skinned with new contexts so recognition cannot masquerade as mastery, and retiring only at L3. Students name the feature themselves ("my dushman is acceleration") and the emotional framing is deliberately playful revenge, not shame: the enemy is the concept, never the child.

Flow: weakest-concepts query → re-skinned questions surface in daily challenge/games → defeat event → retirement at L3 with a badge. **Data:** dushman queue per student, defeat/victory events.

Interest mash-ups VS-45 P1

Lives: question presentation layer | Built: Day 3

Same concept, same difficulty, same scoring — dressed in the student's declared interest (cricket, dance, trains, space). A cricket-mad child meets average speed through run-rates. The math never changes; the willingness does. Interests are chosen by the student in onboarding, changed anytime, and never inferred from anything sensitive.

Flow: interest profile → question skin selection at serve-time → identical scoring path. **Data:** interest tags on students, skin variants on questions.

Absence recovery path VS-20 Core

Lives: Next-Action card | Built: Day 3

After any absence the card offers "Catch-up" — the two most decayed prior concepts plus the missed sessions in compact form, 20 minutes, shame-free copy ("Good to see you. Small steps."). Recovery events are distinct in the trail so heatmaps show a child back on track, not a hole. This tiny flow is one of the product's most humane decisions: most apps punish absence with a wall of backlog, and lose the child precisely when they return.

Flow: absence detected → recovery offer → compact path → recovery event logged → heatmap reflects return. **Data:** attendance/absence signals, recovery sessions, distinct event type.

Test-prep readiness signal VS-47 P1

Lives: profile + parent summary | Built: Day 3

A three-state lamp per upcoming exam — 🔴 Not ready / 🟠 Getting there / 🟢 Ready — computed from mastery across the exam's concept list weighted by prerequisite depth. It replaces the anxious guesswork of "how prepared am I really?" with a number the student can move every night, and gives parents the same truth without any marks drama.

Flow: exam scope defined → readiness query → lamp on profile/parent summary → moves with nightly evidence. **Data:** exam scope table, readiness computation over mastery.

G — Games & Motivation (one bank, five games)

Games arcade VS-93..99 Core

Lives: games tab | Built: Day 4

Five games run on the same question bank and score as concept evidence: **MCQ Rush** (60-second streaks of quick MCQs), **Match-the-Concept** (term ↔ definition pairs against the clock), **Rapid Revision** (the flashcard deck as a self-rated sprint), **Streak Challenge** (bank your current streak's XP or risk it on one more question — a deliberate, tiny taste of downside), and **Boss Level** (the chapter's mixed exam-format boss, gated at ≥70% chapter mastery, whose defeat unlocks the next chapter's preview). Every game is solo — scores live against one's own history, never a class list. The permanent rule from the strategy: **no public leaderboards, ever; motivation comes from streaks, improvement badges and beaten enemies.**

Flow: arcade → game → questions served by band + ladder → score → concept evidence + XP → streak/badge hooks. **Data:** game sessions, per-question outcomes inside games, XP, unlocks.

Streaks with freezes + daily challenge VS-95..96 Core

Lives: student profile + home chip | Built: Day 4

The streak counts any meaningful learning act (session, game, warm-up) — not just full sessions — because 15 honest minutes should not die on a technicality. Every 5 streak days earns a freeze (max 2 stored) that automatically protects an missed day; after a true break, restart is gentle ("welcome back — day 1 starts today") rather than punitive. The 5-minute Daily Challenge feeds warm-ups and Dushman questions, so even a compressed evening moves the needle. Streak logic is server-side and idempotent — one touch per calendar day, ever.

Flow: daily act → idempotent streak touch → freeze accrual/burn → chips and milestone celebrations. **Data:** streak state, freeze ledger, daily challenge completions.

Improvement badges VS-97 Core

Lives: profile badge shelf | Built: Day 4

Badges celebrate trajectories, not talent: "Fixed It — a concept moved L1→L2", "Comeback King — recovered after 3 absent days", "Dushman Slayer — beat your enemy concept", "Boss Buster — cleared the chapter boss". The amber accent colour (the only place it appears in volume) is reserved for this shelf, making achievement visually precious. No badge references other students; every badge is a story the child can retell at dinner.

Flow: evidence events → badge rules engine → award + celebration animation (skippable, reduced-motion aware). **Data:** badge catalog, awards with triggering event IDs.

H — Teacher World

Morning dashboard VS-60 Core

Lives: teacher home | Built: Day 2

One screen answering yesterday's questions before the first bell: completion of last night's sessions, the three reddest heatmap cells, new doubt clusters, submissions waiting for grading, and one suggested action ("2-minute reteach card for 9-B acceleration?"). Nothing here requires data entry — everything is the class's own evidence, aggregated overnight.

Flow: login → dashboard (single query fan-out) → drill into any card. **Data:** aggregate views over sessions, responses, doubts, submissions.

Class × concept heatmap VS-61 Core

Lives: teacher analytics | Built: Day 2

The grid every teacher dreams of: rows = students (or anonymised clusters), columns = concepts, cells = mastery state with the three-colour scale — 🟢 safe / 🟠 unstable / 🔴 critical. Drill into any cell for the underlying signals (confidence pattern, error families, time patterns), and one tap offers "push reteach" — a ready-made repair session to the struggling subset. The heatmap is the teacher's proof that publishing nightly was worth it.

Flow: heatmap → cell drill → signal view → one-tap reteach push → students receive repair path. **Data:** mastery rollups, reteach session records.

Session builder with student preview VS-62 Core

Lives: teacher publish flow | Built: Day 2

Publishing tonight's session takes under three minutes: pick chapter/concepts (auto-suggested from the syllabus progress), attach notes, trimmed video and summary sheet, select the quiz (auto-selected by concept + difficulty mix, editable), set open/close times (18:00–08:00 default), and press preview — which renders exactly what the student will see, in the real player, not a mock. Homework flag toggles the AI into Socratic mode for the session.

Flow: builder → auto-select → edit → student preview → publish → notification to class. **Data:** sessions with sections, resources, quiz refs, timing, flags.

Question bank + AI-drafted questions VS-63 Core

Lives: teacher bank | Built: Day 2

Shared, searchable bank tagged by concept, difficulty and board format. Teachers can ask the AI to draft new questions on a concept ("give me 5 L2 numericals on average speed") — drafts arrive marked as AI-drafted, in a visually distinct style, and enter the bank only after explicit teacher approval. The teacher's own questions are always styled as authoritative; the boundary is never blurred (5.5).

Flow: browse/filter → AI draft request → review → approve/reject → bank entry (provenance kept). **Data:** questions with provenance (teacher vs AI-approved), approvals.

Class space & announcements VS-10 Core

Lives: class workspace feed | Built: Day 2

A quiet, section-scoped notice board: teacher announcements, session publish notices, and gentle class-level celebrations ("9-A crossed a 200-night streak this week"). Section scoping keeps 9-A's feed 9-A's — no school-wide noise, no cross-class comparison content, consistent with the no-leaderboard law.

Flow: teacher posts → class sees notification + feed entry → optional acknowledgement. **Data:** announcements scoped to section, read receipts (aggregate only).

I — Parent World

Weekly progress dashboard VS-73 Core

Lives: parent home | Built: Day 3

Graphs a parent can actually read: effort this week (sessions and minutes per night, as honest bars), mastery trend per subject (the line that matters — up is up), streak health, and the three-state exam readiness lamp. Every graph is generated from the child's real evidence trail, and the plain-language summary underneath explains the graphs in sentences a non-teacher parent can act on. Weekly email digest carries the same truth to the inbox (cron-driven, data-saver friendly).

Flow: parent login → weekly dashboard (aggregates only) → detail per subject → printable report. **Data:** aggregate rollups; never raw responses, never chats (2.4).

"Ask This" conversation prompts VS-75 Core

Lives: parent weekly card | Built: Day 3

One suggested conversation per week, grounded in something real the child did: "Aarav fixed his acceleration graph-reading this week — ask him to show you how he reads one." The prompt turns parents from inspectors into interested witnesses, replaces "how were your marks?" with a question the child can answer proudly, and is deliberately the only prescriptive parenting content in the app.

Flow: week's notable events → prompt generation → card → optional "mark as had". **Data:** prompt templates, delivery/dismiss events (no content snooping).

Notifications & milestones VS-74 Core

Lives: parent + teacher apps | Built: Day 3

Finite, high-signal notifications only — milestone badges, streak milestones, exam readiness transitions, teacher notes — never a stream of "your child opened the app". The notification philosophy mirrors the product's calm: a parent should hear from VidhyaSaathy about twice a week, and every ping should be worth a dinner-table mention. Realtime unread counts for teachers on doubt clusters and submissions.

Flow: events → notification rules → realtime/push → read state. **Data:** notification table, preferences per role.

J — Admin World

School setup wizard + CSV onboarding VS-79..82 Core

Lives: admin console | Built: Day 1-3

Guided setup of school → academic year → classes → sections → subjects → teacher assignments, then bulk import of students and staff from CSV with a column-mapping step, validation preview, and per-row error reporting that names the exact problem (duplicate roll number, bad date format) instead of a generic "import failed". Activation codes are generated per class and exported as printable slips for class teachers to distribute.

Flow: wizard → CSV map → validate preview → import → code export (printable). **Data:** school structure, memberships, import batches with per-row outcomes, codes.

Approvals, permissions & audit VS-83..87 Core

Lives: admin console | Built: Day 3

Parent-link approvals in a queue; a live permission matrix (the one from 2.4, editable only in ways that never violate the privacy wall — that row is hard-locked); and an append-only audit log of every sensitive action (imports, approvals, revocations, resets). The demo-reset script restores the seeded world in one command for repeat presentations — the same world the acceptance script in Part 11 walks.

Flow: queues → approve/revoke → audit entries → exportable log. **Data:** audit events table, immutable.

K — Platform (India layer)

Bilingual + PWA offline VS-88..92 Core

Lives: everywhere | Built: Day 1 (i18n) / Day 4 (PWA)

English and Hindi with full parity — every string ships in both, zero missing keys, switchable anywhere — with the companion speaking natural Hinglish when the student does. The app is a PWA: installable, offline-capable for the current chapter's content ("save for offline"), with an IndexedDB outbox that syncs attempts taken on a flaky connection and marks them clearly as synced-late. Data-saver mode swaps heavy renders for the PDF summaries; Lighthouse targets are ≥80 performance, ≥90 accessibility on a low-end profile, and the whole student loop works on a 2-year-old Android phone on 3G-class networks.

Flow: language switch → instant re-render; offline flag → cached content + outbox → reconnect sync. **Data:** i18n key catalogue (parity-audited), sync log with late-marking.

PART 7

Complete User Flows

Seven flows carry the entire product — from the school's first setup to a game break. Each step shows what the system does (backend) and what the user sees (screen), so nothing is left to imagination or to the AI builder's guesswork.

Flow 1 — School Onboarding (Admin → Everyone)

1

Admin creates the school world

Setup wizard: school profile → academic year → classes 9 & 10 → sections A/B → subjects → teacher-subject-section assignments. Every later permission hangs off these facts.

BACKEND
School, year, class, section, subject and assignment rows written; admin gets owner role; all audit-logged.

USER SEES
A calm wizard with progress indicator, validation on each step, and a final summary card of the created structure.

2

Teachers invited by staff ID + email

CSV or single-entry invite; the email carries a one-time link that verifies staff identity and lets the teacher set a password.

BACKEND
Teacher profile + membership rows; invite token (expiring, single-use); assignment of subject/sections (Sharma Ma'am → Physics 9-A/9-B).

USER SEES
Welcome screen with their classes pre-listed and an empty morning dashboard prompting the first session publish.

3

Students imported; activation codes exported

CSV of roll numbers, names, dates of birth; codes generated per class and exported as printable slips for class teachers to hand out.

BACKEND
Student rows (no credentials yet); activation codes (single-use, expiring) linked to roll numbers; import batch report with per-row outcomes.

USER SEES
Admin sees validation preview before commit and a clean import report naming any problem rows.

4

Student activates with code + roll number + DOB

The student enters the code from the slip, proves identity with roll number and date of birth, sets a password and a recovery question, and lands directly in the student shell.

BACKEND
Credential creation, membership activation, first-login event; language preference recorded.

USER SEES
A two-minute friendly activation, English/Hindi toggle, and a first home with the cloud companion greeting.

5

Parent links via OTP; admin approves

Parent receives a link code through the school, verifies by OTP, and the pending link appears in the admin queue; approval opens the parent world for that child only.

BACKEND
Parent profile, link row (pending → approved), audit entry; aggregates-only read policy attached to the link.

USER SEES
Parent gets a first weekly view with plain-language intro; student is told "your parent can now see your progress — never your chats."

Flow 2 — Student First Login → Diagnostic → Personalised Home

1

Sign-in → class workspace

Roll-number-based identity means the student's world assembles itself: Class 9-A, Physics with Sharma Ma'am, the chapters in syllabus order.

BACKEND
SSR session; one `student_home` RPC returns classes, subjects, chapters, pending items and streak state in a single call.

USER SEES
The bottom-tab student shell with a greeting from the cloud companion and the class workspace one tap away.

2

Chapter open → diagnostic offer

First open of "Motion" offers the 5-question friendly diagnostic ("let's see what you already know — this is not a test").

BACKEND
Diagnostic set drawn from prerequisite chain (distance → average speed → acceleration) + 2 new-concept probes.

USER SEES
Five light questions with confidence buttons and zero marks language; a placement card after ("You're already strong on distance — let's build the rest").

3

Band placement written; home adapts

Per-concept seed scores and the chapter band land in the evidence trail; the home screen's Next-Action card now reflects this specific child.

BACKEND
Mastery seeds, band assignment event, routing profile initialised with neutral explanation-format weights.

USER SEES
A personalised Next-Action card (Priya: repair path; Aarav: acceleration concept) — and no band label anywhere.

Flow 3 — The Daily Learning Session (minute by minute, 18:04)

1

18:04 — Session opens, notification arrives

Sharma Ma'am published at 14:00; the session window opened at 18:00 (closes 08:00). The student gets one quiet notification, not a barrage.

BACKEND
Session state flips to open (server time authoritative); notification dispatched; Next-Action priority recalculated.

USER SEES
The card on home now reads "Tonight's Physics: Acceleration — about 25 min".

2

18:06 — Warm-up offer (if radar flagged decay)

Only when needed: a 2-minute flashcard warm-up on the two faded prerequisite concepts, offered as optional and genuinely short.

BACKEND
Revision radar query over decayed mastery; warm-up completion logged as revision behaviour evidence.

USER SEES
"Quick warm-up? Distance & displacement — fading" with a skip option that is respected without guilt-tripping.

3

18:10 — Concept pages with interactive checkpoints

The Learn station: school notes in the band's variant, checkpoint after each idea, curated video available inline at the teacher's timestamp, "simpler explanation" one tap away.

BACKEND
Note blocks + checkpoint questions served; each checkpoint response scored and logged immediately.

USER SEES
Reading that talks back — a checkpoint card slides in after each idea, and the cloud rests quietly in the corner.

4

18:22 — Quiz with confidence buttons and the ladder

Practice station: questions served per band and the L1/L2/L3 ladder; Sure/Unsure/Guess tapped before each reveal; difficulty moves with performance.

BACKEND
Server-side scoring and error classification per response; ladder transitions logged; bridge explanations attached to any drop.

USER SEES
Questions that get a little harder when they're rolling and kinder when they're not — with an explanation exactly when the ladder drops.

5

18:31 — "Why I got it wrong" summary

Understand station: each miss with its concept tag, error family and one-line fix; the Dushman queue updates silently; ≥60% completes the session, below offers the bridge.

BACKEND
Session completion rule evaluated (all parts + quiz ≥60%); mastery updates written; dushman queue recalculated.

USER SEES
A summary that reads like a coach, not a report card — and the streak chip ticking up.

6

18:34 — Optional game, then done

Four free minutes: MCQ Rush on tonight's concepts (scores count as light evidence), then close. The evening cost 30 minutes and moved three numbers that matter.

BACKEND
Game session logged with concept tags; idempotent streak touch; overnight aggregates refresh for teacher and parent views.

USER SEES
XP, a possible badge, the cloud's small celebration — and a home that has already recalculated tomorrow.

Flow 4 — Doubt to Answer (11 pm, the moment that builds the moat)

1

23:02 — Doubt asked

Stuck on a numerical, the student opens the chat tab and types (or SnapDoubts a photo of the line) — "why is acceleration negative here??"

BACKEND
SnapDoubt extraction → confirmation round-trip; 11-step sequence begins (5.7): auth, rate-limit, context slice, concept map, language + band detection.

USER SEES
The cloud's thinking state — a restrained animation while the answer streams in the student's language and depth.

2

23:04 — Answer at band depth, with tomorrow's hook

Analogy-first for Priya, formula-forward for Aarav; one clarifying question if the doubt was vague; a linked mini-practice queued for tomorrow if the pattern suggests it.

BACKEND
Answer streamed (SSE); interaction logged privately; any promised follow-up written (Dushman/practice queue); budget closed.

USER SEES
An answer that knows where they slipped yesterday — and zero fear of judgement.

3

Morning — the teacher sees the cluster, not the child

"17 doubts on acceleration, 9 on sign convention" — anonymised, clustered, actionable; the student's name appears only if they opted in.

BACKEND
Nightly clustering job; anonymised aggregates to the teacher; opt-in escalations delivered named.

USER SEES
Teacher sees a one-tap "push reteach" on the cluster; tomorrow's class quietly fixes what 11 pm discovered.

Flow 5 — Teacher's Day (from dashboard to reteach)

1

07:40 — Morning dashboard

Last night's completion (9-A: 34/38), three reddest heatmap cells, two doubt clusters, five submissions pending — one glance before the first bell.

BACKEND
Aggregate queries over the evidence trail, computed overnight; realtime counters for doubts and submissions.

USER SEES
A dashboard with one suggested action, not eleven menus.

2

11:20 — Heatmap drill and reteach push

Cell "9-B × acceleration" is critical; drill shows "Sure + wrong" dominance — a misconception, not carelessness; one tap pushes a repair session to the 11 struggling students.

BACKEND
Cell-level signal query; reteach session created and targeted to a student subset; notifications scheduled for 18:00.

USER SEES
Students get a friendly bridge path tonight — none of them ever know they were "the red ones".

3

14:00 — Publish tonight's session (under 3 minutes)

Auto-suggested concepts from syllabus progress; notes attached; quiz auto-selected and edited slightly; open/close times default; student preview verified; publish.

BACKEND
Session row with resources, quiz ref, timing, homework flag; publish notification queued for 18:00.

USER SEES
The real student player in preview — what Sharma Ma'am sees is exactly what Aarav gets.

4

20:30 — Grading queue, gently assisted

Short-answer submissions with AI-suggested marks; she accepts six suggestions in a row and edits two, adding one comment line each — ten minutes, done.

BACKEND
Submissions + AI suggestions fetched; teacher decisions written as authoritative; edits to suggestions logged for calibration.

USER SEES
Students receive marks with a human comment, not a machine verdict.

Flow 6 — Parent's Weekly View (Sunday, 10 minutes)

1

Sunday morning — weekly dashboard

Effort bars for the week, mastery trend lines per subject, streak health, readiness lamp — all generated from the child's own trail, in plain language.

BACKEND
Aggregates-only queries enforced by policy; plain-language summary generated from the same aggregates; weekly email digest mirrors it.

USER SEES
Graphs they can read without a manual, and not one raw answer or chat bubble — by design.

2

The "Ask This" card

"Aarav fixed his acceleration graphs this week — ask him to show you how he reads one." One concrete, warm conversation starter grounded in something real.

BACKEND
Notable-events query → prompt selection; delivery logged, content never snooped.

USER SEES
A dinner-table question their child can answer proudly.

3

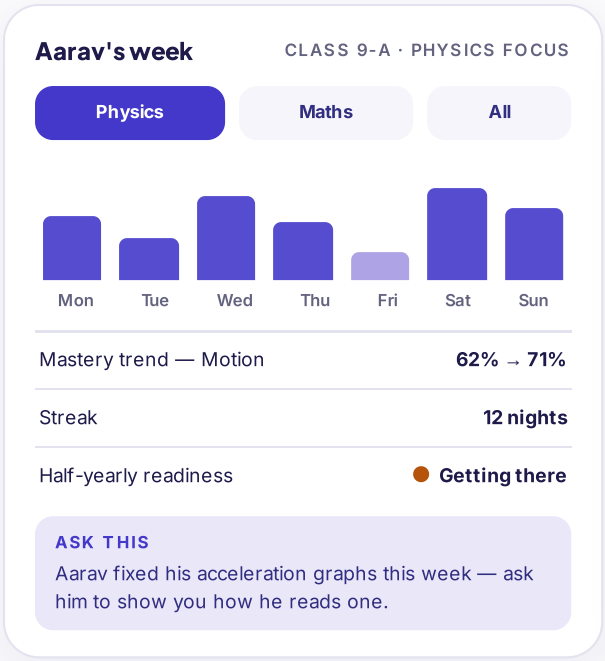
Printable report (when school asks)

A clean one-page PDF: subject trends, effort, readiness lamps, teacher note if any — the PTM handout that used to take an evening to assemble.

BACKEND
Report data package → PDF render (data-saver friendly) → download/share.

USER SEES
A report that matches what the teacher sees — one truth, three audiences.

What the parent dashboard actually looks like — honest bars, one trend line, one lamp, one prompt. Two children, one screen pattern, zero comparison between them: each parent sees their own child's truth, and the plain-language line is honest about effort and about distance still to travel. Note what is absent — no marks table, no class rank, no chat transcripts; absence is a feature here, not a gap.





Flow 7 — Game Break (with the boss gate)

1

Arcade entry — five doors, one bank

All games serve from the same concept-tagged question bank with the same ladder; the games tab shows tonight's recommended door first (warm-up or Dushman). Solo play only — the high score list has exactly one name on it: the player's own history.

BACKEND
Game session bound to student + concepts; per-question outcomes logged as evidence with confidence buttons present here too.

USER SEES
Five cards, time estimates, and the amber Boss Level door — locked until 70% chapter mastery.

2

Streak Challenge — the one taste of risk

Bank current XP or risk it on one more question: a deliberate, bounded lesson in downside, small enough to be fun and never cruel. A wrong risk ends the run with the banked portion intact.

BACKEND
Risk resolution server-side; XP ledger updated; outcome logged with concept tags.

USER SEES
Big numbers, a heartbeat of suspense, and an honest "banked 240 XP" when it ends.

3

Boss Level — the chapter gate

Mixed exam-format boss on the whole chapter; ≥70% chapter mastery to enter, defeat unlocks the next chapter's preview. This is the game that keeps games honest — the boss is the exam's shape, so the fun rehearses the thing that matters.

BACKEND
Gate check on chapter mastery; boss completion writes an unlock event and a badge candidacy.

USER SEES
The cloud's full celebration, the amber "Boss Buster" badge — and a first glimpse of the next chapter.

PART 8

Data & Content Foundation

The app is only as good as the NCERT fuel inside it. This part defines the concept map, the question bank spec, the content quality bar, and the seeded demo world used for every test and demo.

8.1 — The Concept Map (the product's skeleton)

Content lives in a strict hierarchy — **School** → **Year** → **Class** → **Subject** → **Chapter** → **Concept** — and every concept carries prerequisite edges. The prerequisite graph is not decoration: the diagnostic draws from it, the repair paths walk backwards

along it, the study plans sequence by it, and the "concepts are labelled, people are not" rule reads from it. For the prototype, the map is hand-built with teacher review for the seeded chapters, starting with the Motion chain.

Concept	Prerequisites	Why it is in the demo chain
Distance & displacement	Basic algebra, units	Entry concept — separates scalar/vector thinking, common early fork in understanding
Speed & average speed	Distance & displacement	The classic "average speed ≠ mean of speeds" misconception lives here — perfect Error DNA demo
Acceleration	Speed & average speed	Sign convention confusion, graph reading, the highest-doubt concept in the demo data

8.2 — Question Bank Specification

Every question in the bank carries four mandatory tags — **concept, difficulty (L1/L2/L3), board format, and provenance** — plus its options, its correct answer (server-side only), and its error-family hints (which wrong option indicates which confusion). The prototype needs the seeded chapters fully banked, roughly:

Per seeded concept	Diagnostic-grade	Practice pool	Format mix target
Distance & displacement	2 questions	~25 questions	MCQ 40% · Numerical 30% · Assertion-Reason 15% · Case-based 10% · Short 5%
Speed & average speed	2 questions	~25 questions	Same mix, heavier on the average-speed misconception pattern
Acceleration	1 probe + 2 prerequisite pulls	~30 questions	Same mix, heavier on sign convention and graph reading

Quality bar for every question: unambiguous stem, exactly one defensibly correct answer, distractors that *mean something* (each maps to a real confusion), bilingual-capable (stem at minimum in English and Hindi), and no trick questions — difficulty must come from thinking, not wording. Ten excellent questions beat thirty mediocre ones, because every bad question corrupts the evidence trail the whole moat stands on.

8.3 — Content Pipeline for the Prototype

- **Teacher-authored notes** for the three seeded concepts (Sharma Ma'am's actual emphasis, ~1 page each, with 2-3 checkpoint questions embedded per concept).
- **Simple and concise variants** of each note for Band-1 and Band-4 serving — same truth, different depth (analogies and shortest sentences vs. compact advanced form).
- **Summary one-pager PDF** for the Motion chapter — definitions, core formulas, two classic exam patterns, mini concept map.
- **Flashcard decks** — 8-12 cards per concept, fronts as terms/mini-questions, backs in one breath.
- **Two curated YouTube resources** with teacher-set timestamps and one-line reasons (real links, trimmed ranges).
- **Hindi translations** of every student-facing string and every content piece — full parity, audited to zero missing keys.

8.4 — The Seeded Demo World

Every test and demo runs on one deterministic, re-loadable world (the reset script restores it in one command): Green Valley School (CBSE); Class 9-A and 9-B; Sharma Ma'am (Physics) plus one maths teacher; the Motion chapter with its three-concept chain; tonight's session published at 14:00, open 18:00-08:00; and three student archetypes whose evidence makes every screen demonstrable — **Aarav** (strong: 5/5 diagnostic, L3-L4 chain, 12-night streak), **Priya** (struggling: fragile average-speed prerequisite, mid-repair journey, warm-tone AI history), and **Rohan** (absent 3 days: decayed mastery, recovery path pending). Their parents are linked and approved; the badge, game and streak catalogs are seeded; and the doubt history includes a genuine acceleration cluster (17 doubts, 9 on sign convention) so the teacher's morning dashboard tells a true story on first load.

PART 9

The 4-Day Build Plan

From the current static-HTML build to a working, adaptive, demo-ready app in four intense days. The ten Build Prompt phases map onto the days exactly as written below — one phase at a time, question gate honoured, checklist verified, no stacking.

9.1 — Day 0 (Evening Before) — Preparation

Four days of building fail for logistics reasons more often than technical ones, so the evening before is a real work block: Supabase project created and keys stored in env vars (never committed); the Git repo audited and the existing static screens inventoried (the Build Prompts' first instruction); the seeded NCERT content pack (notes, variants, summary sheet, flashcards, ~80 banked questions, Hindi translations) staged and ready to load; demo accounts decided (admin, Sharma Ma'am, Aarav, Priya, Rohan, two parents); and the AI provider key provisioned with a spend cap. Content readiness is the difference between a Day 2 that demos learning and a Day 2 that demos skeletons.

9.2 — The Four Days

Day 1 — Foundation

Build Prompts 01 + 02 + 03 · Database, Entry System, App Shell

Morning — Phase 01
Full schema migration (8 groups: identity, academic structure with the concept map, content, assessment with server-side keys, evidence/progress, AI/doubts, communication/motivation, ops) + RLS policy matrix + core RPCs (submit_quiz, touch_streak) + storage buckets + demo seed.

Afternoon — Phase 02
School-anchored auth end-to-end: admin wizard, teacher invite, student activation (code + roll no + DOB), parent OTP link, SSR sessions, role guards, forgot password.

Evening — Phase 03
Scholar Indigo token system as the single source of truth, component kit in all states, the four role shells, notification infrastructure, bilingual i18n skeleton, "continue where you left off".

EXIT DEMO (MUST PASS TO SLEEP)

All four roles sign in with real credentials and land in their real shells; a student's class workspace reads real seeded data; no mock logins anywhere; RLS proven by a rejected cross-role query.

Day 2 — The Learning Loop

Build Prompts 04 + 05 · Student Core Loop, Teacher World

Morning — Phase 04
student_home single-call RPC; session engine (open/close windows, resume pointer, ≥60% completion); quiz engine server-scored; confidence buttons on every question; "Why I got it wrong" with concept + error family; per-concept mastery view; the Next-Action card.

Afternoon — Phase 05
Teacher morning dashboard; class × concept heatmap with drill and one-tap reteach; session builder with real student preview; question bank with auto-selection; class space announcements.

Evening — buffer
Fix the day's checklist failures; run Flow 3 (daily session) manually as Aarav; seed deepens if time allows.

EXIT DEMO (MUST PASS TO SLEEP)

Sharma Ma'am publishes tonight's session at 14:00 in under 3 minutes; at 18:05 Aarav completes it end-to-end — warm-up, concept page with checkpoints, quiz with confidence buttons, "why wrong" summary — and her heatmap updates from his real answers.

Day 3 — The Intelligence

Build Prompts 06 + 07 + 08 · Parent/Admin, AI Tutor, Personalisation

Morning — Phase 07
Cloud companion chat via the 11-step Edge Function sequence (SSE streaming, server-side keys, rate limits + budget); band-depth answers; Socratic homework mode; honesty layer; SnapDoubt with confirmation; doubt clustering to the teacher inbox.

Afternoon — Phase 08
Band routing live in every serving surface (note variants, quiz entry, ladder); Error DNA repair paths; promotion/demotion logging; revision radar warm-ups; absence recovery path; Dushman queue; study plans; frustration tone; interest mash-ups.

Evening — Phase 06
Parent weekly dashboard with real graphs from the trail; "Ask This" cards; weekly email digest cron; admin CSV import, code export, approvals, audit log, permission matrix.

EXIT DEMO (MUST PASS TO SLEEP)

Priya and Aarav ask the same doubt and get visibly different-depth answers; Priya's repair path and Rohan's catch-up card both work; a parent opens a dashboard whose graphs moved because of last night's real session; the privacy wall is proven by a rejected parent query on chats.

Day 4 — Delight & Launch

Build Prompts 09 + 10 · Games & Motivation, India Layer, QA, Launch

Morning — Phase 09
The five games on the shared bank (MCQ Rush, Match-the-Concept, Rapid Revision, Streak Challenge, Boss Level with the 70% gate); streaks with freezes; improvement badges; daily 5-minute challenge; every game writes concept evidence.

Afternoon — Phase 10
PWA + service worker (save for offline, outbox sync with late-marking); data-saver mode; full Hindi audit to zero missing keys; low-end Lighthouse pass (≥80 perf / ≥90 a11y); edge-case sweep; full-day acceptance script (11.2) executed for real.

Evening — launch block
Two fix rounds for anything the acceptance script surfaced; deploy; runbook written; demo-reset script verified; the Part 1.5 sentence is demonstrated live on a real phone.

EXIT DEMO (THIS IS THE DEMO)

The complete Part 11 acceptance script passes end-to-end on a deployed URL — admin 07:00 through next-morning heatmap — and the three archetypes have three visibly different evenings.

9.3 — Daily Rhythm (the discipline that makes 4 days possible)

- **One phase at a time, fully.** The AI builder receives a phase, answers its question gate, builds, verifies its own checklist, reports — then and only then the next phase. Stacking phases is how static-HTML chaos was born; do not revive it.
- **Test as the user, not as the developer.** Every exit demo is walked on a real (preferably low-end) phone with the seeded accounts, in both languages at least once a day.
- **No regressions.** Each day ends with a five-minute regression pass of the previous days' exit demos. A feature that breaks yesterday's demo is not a new feature; it is a bug.
- **Honest triage over silent scope creep.** Anything slipping past its day goes through the 9.4 table explicitly — in the open, recorded, never by quietly shipping it half-broken.

9.4 — If Behind: Triage Table (in this order, no exceptions)

Slipping item	Day-4 action	Why it is safe to defer
Interest mash-ups, readiness lamp refinements	Defer to post-launch backlog	Nice differentiation, not load-bearing for the demo sentence
SnapDoubt photo pipeline	Ship typed doubts only	The chat already carries the doubt; the camera is sugar
Streak Challenge game	Ship four games, not five	The arcade story survives with four doors; Streak Challenge is the most deferrable

Weekly email digest	In-app parent view only	The dashboard proves the graphs; cron polish can follow
PWA offline depth	Install + cache shell only	Full outbox sync is Day-4-late acceptable; online demo works regardless
● Never defer	Server-side scoring · RLS · confidence buttons · band routing · teacher heatmap · parent aggregates · the evidence trail	These ARE the product — deferring any of them means the demo shows a mock

PART 10

The Complete Readiness Checklist

The master tracker — every single thing that must be true, from sign-up to final testing, in thirteen groups. Tick items as they are verified in the real app (not as they are "probably done"). This section is the file you update daily.

Rules of engagement for this tracker: an item is ticked only when it has been **observed working in the deployed app** by a human, in the role it belongs to. "The code is written", "the AI says it did it", and "it worked yesterday" do not count. When an item fails, it goes back to its Build Prompt phase with a one-line note — the tracker never lies to make anyone feel better, because the demo (11.2) will not.

10.1 Backend & Database

16 ITEMS

- [ ] Full schema migrated: all 8 table groups exist with correct relations
- [ ] RLS enabled on every table — no policy gaps found by audit query
- [ ] **Answer keys never reach the student client** (network trace verified)
- [ ] touch_streak is idempotent (double-tap tested, single increment)
- [ ] Storage buckets private with signed-URL access only
- [ ] Session open/close enforced by server time (client clock spoof ignored)
- [ ] No secret keys in the repo (grep clean); env vars only
- [ ] Audit log captures imports, approvals, revocations, resets
- [ ] Concept prerequisite map seeded (Motion chain verified)
- [ ] Cross-role read attempt rejected in logs (e.g. parent → student chats)
- [ ] submit_quiz RPC scores server-side and writes per-question evidence
- [ ] Evidence trail is append-only; corrections create new events
- [ ] Demo seed loads in one command; reset script restores it exactly
- [ ] Migrations re-runnable on a fresh project without error
- [ ] Indexes on hot paths (student_home query under 300 ms on seed)
- [ ] Nightly aggregates job runs (heatmap/parent data fresh by morning)

10.2 Auth & Onboarding

14 ITEMS

- [ ] Admin wizard creates school → year → classes → sections → subjects
- [ ] Teacher invite (staff ID + email) → invite link → password set → lands in shell
- [ ] Student activation: code + roll no + DOB → password → student shell
- [ ] Wrong roll no / DOB / expired code → named, friendly error (no leak)
- [ ] Codes are single-use (second attempt rejected) and expiring
- [ ] Parent link: OTP → pending → admin approval → parent world for that child only
- [ ] Parent link revocation works and is audit-logged
- [ ] Forgot-password flow completes by email for every role
- [ ] SSR session: refresh keeps role; logout kills session everywhere
- [ ] Role guard redirects wrong-role URL access (student → /admin bounces)
- [ ] Auth screens fully bilingual (EN/HI switch live)
- [ ] First student login greets via cloud companion; zero empty states
- [ ] CSV student import: validation preview + per-row error naming
- [ ] Activation-code export produces printable slips per class

10.3 Student Experience

18 ITEMS

- [ ] Home loads in one RPC call; Next-Action card is the visual anchor
- [ ] Next-Action priorities correct (unfinished > due > revision > new > games)
- [ ] Diagnostic offered on first chapter open; 5 questions; friendly framing
- [ ] Diagnostic result routes content (Priya ≠ Aarav verified side-by-side)
- [ ] Session player: resume mid-state works after app kill
- [ ] Concept page: school notes render with formatting + Hindi
- [ ] Interactive checkpoints fire, score, and explain inline
- [ ] "Explain simpler" swaps note variant for the same idea
- [ ] Curated YouTube plays inline at teacher's timestamp range
- [ ] Summary one-pager PDF downloads/prints cleanly
- [ ] Flashcards flip; self-ratings (knew/almost/forgot) logged
- [ ] **Confidence buttons on every question** (quiz + games)
- [ ] Quiz summary shows "why wrong" per miss with fix line
- [ ] ≥60% completion rule enforced; below → bridge path offered
- [ ] Mastery map in profile shows per-concept L1-L4, friendly words only
- [ ] No band labels, ranks, or classmate scores anywhere (audited)
- [ ] Absent student sees shame-free "Catch-up" card (Rohan verified)
- [ ] Chapter history persists — "forgot acceleration?" reopens in March

10.4 Teacher Experience

14 ITEMS

- [ ] Morning dashboard shows completion, red cells, clusters, queue, one action
- [ ] Heatmap grid correct vs seed; colours match safe/unstable/critical rules
- [ ] Cell drill shows confidence pattern + error families for that student × concept
- [ ] One-tap reteach pushes repair session to the struggling subset only
- [ ] Session builder: auto-suggested concepts, attach notes/video/summary
- [ ] Quiz auto-selection by concept + difficulty mix; editable
- [ ] **Student preview renders the real player**, not a mock
- [ ] Publish takes < 3 minutes timed, twice in a row
- [ ] Publish notification reaches class at 18:00 open (not before)
- [ ] Question bank filters by concept/difficulty/format; provenance visible
- [ ] AI-drafted questions marked visually; approve/reject both work
- [ ] Grading queue: AI suggestion, teacher accept/edit, comment line, returned
- [ ] Doubt clusters anonymised; opt-in named escalation delivers correctly
- [ ] Subject/section scoping: Sharma Ma'am cannot touch 9-C or Maths rows

10.5 Parent Experience

10 ITEMS

- [ ] Weekly dashboard graphs render from the child's real trail
- [ ] Plain-language summary reads like a person, updates weekly
- [ ] **Privacy wall proven**: no chats, doubts, or raw answers reachable
- [ ] Notifications only for milestones/readiness shifts (no spam)
- [ ] Weekly email digest delivered (or triaged per 9.4 with reason recorded)
- [ ] Effort bars, mastery trend, streak, readiness lamp all match seed truth
- [ ] "Ask This" card grounded in a real event of that child
- [ ] Multi-child parent switches children; only linked ones visible
- [ ] Printable one-page report generates and matches teacher's truth
- [ ] Parent screens bilingual; no jargon without plain-language twin

10.6 Admin Experience

10 ITEMS

- [ ] Setup wizard completes without reference docs (usability test)
- [ ] Activation codes generated per class; export prints correctly
- [ ] Permission matrix page matches Part 2.4 exactly
- [ ] Audit log lists sensitive actions with actor, time, target
- [ ] Admin cannot read student AI chats (matrix + query both verified)
- [ ] CSV import: happy path + one bad row → named error, others commit
- [ ] Parent-link approval queue works; revoke cuts access immediately
- [ ] Privacy-wall rows are hard-locked in the matrix UI (cannot be toggled)
- [ ] Demo-reset script restores the seeded world verbatim
- [ ] Admin console navigation limited to admin role

10.7 AI & Personalisation

16 ITEMS

- [ ] Chat runs the full 11-step server sequence; keys never client-side
- [ ] Same doubt, different depth: Priya vs Aarav outputs differ appropriately
- [ ] Homework mode refuses direct answers; guides Socratically
- [ ] No fabricated citations in 20-probe adversarial spot-check
- [ ] SnapDoubt: photo → extraction → confirmation → doubt (or triaged, 9.4)
- [ ] Error DNA families drive four visibly different repair paths
- [ ] Band promotion event logged and visible to teacher ("moved 25-50 → 50-75")
- [ ] Dushman queue targets 3 weakest concepts; retires at L3
- [ ] Answers stream (SSE) with the cloud's thinking state
- [ ] English, Hindi, and natural Hinglish all answered in-language
- [ ] Honesty path: unsure questions get "I don't know" + escalation offer
- [ ] Rate limit + token budget enforced (burst test rejected cleanly)
- [ ] Doubt clustering job produces "17 on acceleration" from seed truth
- [ ] Ladder: 3-up / 2-down rule observed live with bridge text on drops
- [ ] Revision radar warm-up appears exactly when decay is seeded
- [ ] Study plan generates realistic nights from trail; shrinks after a miss

10.8 Games & Motivation

12 ITEMS

- [ ] All five games (or four per 9.4) playable end-to-end on low-end phone
- [ ] Every game scores against concepts and writes evidence rows
- [ ] Confidence buttons present inside games too
- [ ] Ladder applies inside games (difficulty moves with performance)
- [ ] Boss Level locked below 70% chapter mastery (negative test done)
- [ ] Boss defeat unlocks next-chapter preview + badge candidacy
- [ ] Streak accrues from any learning act; freeze burns on missed day
- [ ] Freeze rules: 1 per 5 days, max 2 stored (boundary tested)
- [ ] Daily 5-minute challenge serves warm-up/Dushman content
- [ ] Badges award on real trajectories (L1→L2, comeback, dushman, boss)
- [ ] **No leaderboard, no classmate comparison anywhere** (audited)
- [ ] Celebrations skippable; reduced-motion honoured in games too

10.9 Content & NCERT Data

12 ITEMS

- [ ] Motion chain: 3 concepts + prerequisites seeded and correct
- [ ] ~80 banked questions meet the four-tag rule (concept/L/format/provenance)
- [ ] Distractors map to real confusions (spot-checked with a teacher)
- [ ] Notes for 3 concepts in 3 variants (simple/standard/concise)
- [ ] Interactive checkpoints embedded (2-3 per concept) and scored
- [ ] Summary one-pager exists for Motion; teacher-approved trail present
- [ ] Flashcard decks: 8-12 cards × 3 concepts; content correct
- [ ] Two curated videos with real timestamps + one-line reasons
- [ ] Three archetypes seeded exactly per Part 8.4 (strong/struggling/absent)
- [ ] Doubt cluster history seeded (17 acceleration / 9 sign convention)
- [ ] Badge/game/streak catalogs seeded with icons + bilingual names
- [ ] Scope respected: NCERT 9-10 only, nothing else crept in

10.10 Design System & Polish

12 ITEMS

- [ ] Scholar Indigo tokens are the single source (no hard-coded colours)
- [ ] Hero gradient appears ONLY on hero/auth surfaces (audit clean)
- [ ] Amber accent < 5% of visual area (badges/achievement only)
- [ ] Buttons/inputs/cards/modals all have loading, empty, error states
- [ ] Skeletons on every slow surface; no layout jump on load
- [ ] Error + retry pattern with friendly copy everywhere it can fail
- [ ] 4-pt spacing grid respected (spot audit of 10 screens)
- [ ] Touch targets ≥ 44px on all primary actions
- [ ] Cloud mascot states wired (thinking in chat, rest during solving)
- [ ] No mascot animation during question solving (verified visually)
- [ ] Night variant readable (dark mode spot check)
- [ ] No lorem ipsum / placeholder text anywhere (grep + visual audit)

10.11 Performance, PWA & India Layer

12 ITEMS

- [ ] Lighthouse mobile: performance ≥ 80, accessibility ≥ 90
- [ ] PWA installable; app shell caches; icon + splash correct
- [ ] Offline attempts land in outbox; sync on reconnect with late-marking
- [ ] Hindi audit: zero missing keys (automated parity check green)
- [ ] Hinglish flows naturally in companion chat (native review)
- [ ] Battery/memory sane after 15-minute session on low-end device
- [ ] Student loop usable on 2-year-old Android over 3G-class network
- [ ] "Save for offline" caches current chapter; airplane-mode read works
- [ ] Data-saver mode swaps heavy renders for PDF summaries
- [ ] Language switch instant, in place, state preserved
- [ ] No giant images on student paths; everything lazy or compressed
- [ ] Reduced-motion setting honoured globally (OS-level test)

10.12 Testing & Acceptance

14 ITEMS

- [ ] Full-day acceptance script (11.2) executed once, end-to-end, on record
- [ ] Edge-case sweep (11.3) done: every listed case handled gracefully
- [ ] Permission matrix negative tests all rejected (parent/admin walls)
- [ ] Streak idempotency across timezones/midnight edge verified
- [ ] Regression pass of Days 1-3 exit demos green after Day 4 changes
- [ ] Fresh-eyes walkthrough: someone who did not build it completes a session unaided
- [ ] Demo-reset verified AFTER final fixes (the world is repeatable)
- [ ] Three-archetype evening visibly different (recorded side-by-side)
- [ ] Both languages walked through the full student loop
- [ ] Rapid double-submit / flaky-network quiz submissions stay consistent
- [ ] AI adversarial probes: no citations fabricated, no fear/shame tone
- [ ] Fix round 1 items closed; fix round 2 items closed (or triaged)
- [ ] All Part 10 sections ticked honestly; failures carry one-line notes
- [ ] The Part 1.5 sentence demonstrated live on a real phone — pass

10.13 Launch & Demo Readiness

10 ITEMS

- [ ] Deployed URL live; HTTPS; env vars set in hosting (not local-only)
- [ ] Runbook written: deploy, reset, seed, rotate keys, known limits
- [ ] Demo accounts + passwords listed in one secure place for presenters
- [ ] AI spend cap + rate limits set for demo-day safety
- [ ] Team knows the triage table by heart (or has it printed)
- [ ] PDF metadata + analytics/error reporting basics on (or triaged)
- [ ] Demo script rehearsed once against the deployed world, timed
- [ ] Fallback demo plan if venue network dies (offline-cached device)
- [ ] Post-launch backlog (Part 12.3) written down where the team will see it
- [ ] Someone celebrated — four days of this deserves it

PART 11

Quality Gates & Final Testing

The global definition of done, the full-day acceptance script that proves the product, the edge-case sweep, and the two fix rounds that turn "mostly working" into demo-ready.

11.1 — Global Definition of Done (applies to every phase, every day)

- **Real data, real flows.** Every list, score, graph and notification is computed from the database on a real role's session — no placeholder JSON, no mock logins, no "will wire later".
- **Both languages.** Anything a user can read exists in English and Hindi with zero missing keys at the moment it ships.
- **All states designed.** Loading, empty, error + retry, and permission-denied states exist for every async surface — the skeleton is part of the feature.
- **Security proven, not assumed.** Each new capability gets one positive test (right role succeeds) and one negative test (wrong role rejected, in the database, not just hidden in UI).
- **Evidence written.** If the feature touches learning, it writes to the trail with the right event shape — or it is a screen, not a feature.
- **No regressions.** Previous exit demos still pass before the phase is called done.

11.2 — The Full-Day Acceptance Script (run this exactly, on the record)

Time	Actor	What must happen (observed, not assumed)
07:00	Admin	Signs in; dashboard healthy; approves one pending parent link; exports a fresh activation-code batch; audit log shows both actions
09:30	Teacher	Morning dashboard shows last night's completion and the seeded doubt cluster; drills the acceleration cell; pushes one reteach to the struggling subset
14:00	Teacher	Publishes tonight's Motion session (timed: under 3 minutes) with notes, checkpoint-embedded concept page, summary PDF, curated video, and a 5-question quiz
18:00	System	Session opens on server time; notifications land quietly; Next-Action cards update
18:05	Aarav	Skips warm-up (radar clear), concise variant, L3 quiz, 4/5 "Sure", one method-error fix, Dushman queued, MCQ Rush for fun, streak ticks to 13
18:20	Priya	Repair path first (bus analogy), simplest variant, Hindi labels, L1 quiz 3/5 with one discounted "Guess", warm companion tone, bridge scheduled
20:30	Rohan	Catch-up card, 20 minutes, shame-free copy, recovery event logged
22:45	Priya	Asks the companion one real doubt; streamed answer at her depth; doubt joins the anonymous cluster
21:30	Parent	Opens weekly dashboard; graphs moved because of tonight's real events; "Ask This" card reflects a true milestone; no chats anywhere in sight
Next 07:30	Teacher	Heatmap cells greener where the reteach landed; completion counts correct; grading queue behaves
Anytime	Tester	Language switch mid-flow; offline save + outbox sync; low-end device walkthrough; permission-wall negative probes all rejected

> **The pass condition:** every row observed by a human who did not build it, in one continuous run, on the deployed URL. Screenshots or it did not happen — they go next to the Part 10 ticks.

11.3 — Edge-Case Sweep (before calling anything "done")

Situation	Required behaviour
Student kills app mid-quiz	Resume lands on the same question; no double-submit; timer state sane
Two tabs, same quiz	One authoritative attempt; second tab gets a gentle "already in progress"
Session window missed entirely	No shame copy; item folds into tomorrow's plan; parent sees honest effort line
Midnight streak boundary	Idempotent touch survives; timezone set by school, not device
Flaky network during submit	Outbox holds the attempt; sync marks it late; teacher sees "synced late" tag, not silence
AI provider down / timeout	Companion shows honest error state + retry; queue suggestion; never a fake answer
Teacher publishes empty session	Validation blocks publish naming exactly what is missing
Student activates with wrong DOB thrice	Friendly lock with "ask your class teacher" path; no account lockout spiral
Parent link revoked mid-week	Parent world closes immediately; audit entry; child experience unchanged
Hindi toggle mid-session	In-place switch; no lost progress; question text language matches choice
Rapid session re-opens	Completion rule still enforced; no XP/badge double-award

11.4 — The Two Fix Rounds

After the acceptance script, the calendar holds exactly two fix rounds — Round 1 for everything that broke the script or the checklist (severity first: data integrity, security, then flows, then polish), and Round 2 for the fallout of Round 1 plus the triage decisions from 9.4 made real (deferred items moved to the backlog properly, half-built features either finished or hidden, never left dangling). After Round 2 the build freezes: the remaining hours go to rehearsal and rest, because a tired demo of a working app loses to a fresh demo of a slightly-less-working one, every single time.

PART 12

The Moat — Why This Compounds

What is deliberately never built, what the flywheel becomes after launch, and the order of the backlog that follows.

12.1 — What We Deliberately Never Build

Strategy is what you say no to, so the cuts are recorded as product law, not as TODOs in disguise: **no public leaderboards or any classmate-vs-classmate comparison** (motivation here is personal — streaks, badges, beaten enemies); no coaching-institute integrations or ERP/fees/transport modules (the school's learning world is the boundary); no camera or microphone emotion detection (cut forever — creepy, unreliable, and it would poison the trust the privacy wall exists to build); no AR/VR in this horizon. Every one of these was weighed and refused with reasons; when someone requests them in month three, this section is the answer.

12.2 — What the Flywheel Becomes

After the four days, the app is not finished — it is *armed*. Every night it learns the school a little better: which concepts produce which confusions in this section, which explanations rescue which kinds of errors, which students are quietly afraid of which chapters. Six months of that is a dataset no competitor can buy and a rapport no generic app can fake — teachers' reteaches get pre-suggested, parents' conversations get grounded, students' plans get personal. The features were never the moat; the **compounding knowledge of each child, inside one trusted school** is. That is why Part 3's evidence trail was

built before any beautiful screen, and why every new idea must answer "which station of the loop does this serve?" before it earns an hour.

12.3 — Post-Launch Backlog (in priority order, after real usage)

1. **Signature AI suite completion** — the deeper doubt tooling and assistive surfaces that were consciously deferred so the core loop shipped honest and fast.
2. **Recordings pipeline** — class-recording segments with school permission controls, closing the last content gap for absent students.
3. **PIN profiles for shared phones** — because many households share one device, and the streak must survive it.
4. **WhatsApp fallback notifications** — meeting parents where they already live, with the same two-notices-a-week discipline.
5. **Advanced teacher tooling** — deeper analytics exports, co-teaching views, department rollups.
6. **Second school** — only when the first one runs itself for a month. Multi-tenancy before repeatability is how pilots die.

12.4 — The Sentence to Remember

> "VidhyaSaathy removes the chaos: it tests before it teaches, adapts to the four kinds of learner in every classroom, answers doubts at 11 pm without judgement, and turns every answer into evidence that helps the teacher tomorrow and the parent on Sunday. The app is the moat — and it starts compounding the first night it is used."
>
> PRODUCT MASTER FILE · CLOSING DEFINITION

Print that sentence, tape it to the monitor, and check the build against it every evening. When all twelve parts of this file are true in the deployed app — every flow working, every checklist ticked honestly, every band routing a real child differently — VidhyaSaathy stops being beautiful HTML pages and becomes what it was always meant to be: the calm, honest learning companion that an Indian classroom can finally rely on.

VIDHYASAATHY · PRODUCT MASTER FILE

Small steps make big progress.

One loop, four bands, four worlds, four days — and every answer a child gives becomes the evidence that makes tomorrow's lesson smarter. That is the product. That is the moat.

12 parts | 7 flows | 5 study surfaces | 170 checks | 4 days

VERSION 1.0 — DEFINITIVE

SCOPE — NCERT CLASS 9 & 10

COMPANION — BUILD PROMPTS 01-10

OWNER — FOUNDING TEAM

